

**INTERNATIONAL CYBERSECURITY AND
DIGITAL FORENSICS ACADEMY**

GlobalHealth Connect (GHC)

**Information Security Governance
Laboratory Report**

Consolidated Evidence Bundles for Tasks 1-5

Student Name	Kefilwe Matake
Registration Number	C11/26/CGRCE/17575
Course	GRC102 – Information Security Governance
Assignment	Week 1 Practical Laboratory
Date	05–11 September 2026

Executive Summary

GlobalHealth Connect has outgrown an informal security model. The organisation has expanded rapidly, completed two acquisitions and experienced a contained data-leakage near-miss. In a health-technology environment, that combination creates a governance risk because patient information, cloud services, regulatory obligations, uptime and customer trust all depend on security being owned and managed consistently.

The overall finding is that GHC currently operates between Level 1 - Ad Hoc and Level 2 - Initial maturity. The recommended direction is to formalise security governance through an approved charter, a dedicated governance function, a cross-functional Security Steering Committee, Board-level reporting and a 12-18 month roadmap to reach Level 3 - Defined maturity across all governance domains. This approach follows recognised governance principles that cybersecurity should be understood, assessed, prioritised and communicated in a way that supports organisational risk management (NIST, 2024).

The report is intentionally practical. It does not treat security as only a technical function. It positions information security as a business governance capability that supports market expansion, operational resilience, customer trust, secure innovation and regulatory excellence.

Task 1: Governance Blueprint

Executive request: The CEO requires a formal security governance structure with clear roles, reporting relationships and accountability following the recent near-miss.

Evidence Item 1.1 - Current-State Governance Gap Assessment

GHC has reached a point where an informal security approach is no longer enough. The current model creates gaps in accountability, executive visibility and consistency. The near-miss should therefore be treated as a warning sign that security governance must become a formal management discipline rather than a set of activities performed when issues arise.

Governance gap	Current weakness	Why this matters
No formal security governance structure	Security appears to be handled informally rather than through an approved governance model.	GHC may struggle to show who owns security decisions and how risks are being managed.
Security is too dependent on IT Operations	John Smith, the IT Manager, manages infrastructure and also performs some security activities.	This creates over-reliance on one role and does not provide enough independence between operations and governance.
Lack of clear accountability	There is no clearly defined owner for security governance, policy oversight, risk reporting or incident readiness.	After a serious incident, it may be unclear who is responsible for decisions, escalation and reporting.
Weak Board-level visibility	The Board wants meaningful metrics, but there is no evidence of regular governance reporting.	Executives may not have enough information to understand security risk in business terms.
Security is not fully cross-functional	HR, Finance, Development, Compliance and IT all handle sensitive processes, but responsibilities are not formally joined up.	Security may be treated as an IT problem rather than an organisation-wide responsibility.
Inconsistent controls after acquisitions	The two acquisitions may have introduced different systems, processes and security practices.	This increases the risk of control gaps and weak integration of acquired environments.
Incident response is not formally governed	The recent near-miss was contained, but there is no clear evidence of formal incident response planning.	A future incident may cause regulatory, reputational, financial or operational damage if roles and escalation paths are unclear.
Risk management is not embedded into business decisions	Security risk does not appear to be formally reviewed across business, technology and compliance teams.	Risks may only be identified late, after systems are live or decisions have already been made.

Evidence Item 1.2 - Proposed Security Governance Organisational Chart

GHC should establish a dedicated Information Security Governance Function led by the Director of Information Security Governance. The role should report to the CEO and have direct reporting access to the Board or Risk and Audit Committee for material risks, incidents and maturity updates. This is consistent with governance expectations that roles, responsibilities and risk-management communication must be defined and visible (NIST, 2024).

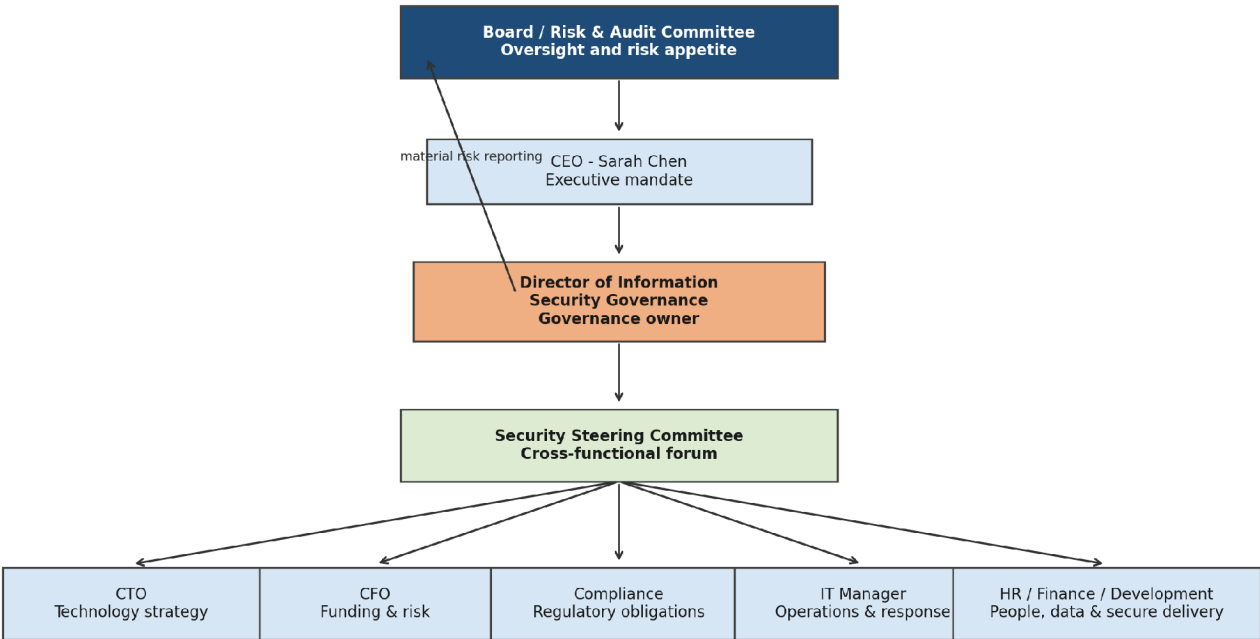


Figure 1. Proposed security governance organisational structure.

Evidence Item 1.3 - RACI Matrix for Security Governance Activities

RACI key: R = Responsible; A = Accountable; C = Consulted; I = Informed.

Task 1 - Evidence Item 1.3: RACI Matrix

Governance activity	Board	CEO	DISG	IT Mgr	Senior Dev	Compliance	HR	Finance/Data	SSC
Approve Information Security Charter	A	C	R	I	I	C	I	I	C
Approve enterprise security policies	A	C	R	C	C	C	C	C	C
Maintain security policy framework	I	I	A/R	C	C	C	C	C	C
Quarterly security risk review	I	A	R	C	C	C	C	C	C
Incident response planning	I	A	R	C	C	C	C	C	C
Incident response execution	I	A	C	R	C	C	C	C	I
Security awareness and role-based training	I	C	A	C	C	C	R	C	C
Secure development governance	I	C	A	C	R	C	I	I	C
Access control governance, including onboarding and offboarding	I	C	A	R	I	C	R	C	C
Board-level security metrics and reporting	A	C	R	C	C	C	C	C	C

Evidence Item 1.4 - Governance Rationale

Accountability: The proposed structure creates a clear governance owner through the Director of Information Security Governance, while the CEO remains accountable for giving the function authority and support. John Smith remains important for operational security, but he is no longer the only person carrying security accountability for the whole organisation.

Transparency: The structure creates a formal reporting path from the governance function to the CEO and Board. Risks, incidents, exceptions and remediation progress can be reported consistently and in business language.

Business alignment: The Security Steering Committee makes security a shared business responsibility. Development, HR, Finance, Compliance and IT all influence security outcomes, so they must participate in the governance forum.

Risk management: The proposed model moves GHC away from waiting for incidents and towards regular risk reviews, policy oversight, incident planning and Board-level reporting.

Task 2: Information Security Charter

Executive request: The CFO and Board require a formal mandate for the information security programme.

Evidence Item 2.1 - Completed GHC Information Security Charter

Charter status

This Charter is designed for executive approval and organisation-wide reference. It creates the mandate, authority, responsibilities and reporting expectations for the Information Security Programme.

1. Purpose

The purpose of the GlobalHealth Connect Information Security Programme is to protect the organisation's patient information, business systems, cloud platforms, financial information, intellectual property and reputation.

GHC has grown rapidly through new business expansion and acquisitions. The recent data-leakage near-miss shows that security can no longer be managed informally or only as a technical activity. The programme therefore exists to provide a formal, business-aligned approach to managing information security risk across the organisation.

The programme supports GHC's 2026 strategic goals by helping the business grow safely, maintain customer trust, improve operational resilience, support secure innovation and meet applicable healthcare and data-protection obligations. This aligns with the ISO/IEC 27001 view that information security management should apply a risk process adapted to the organisation's size and needs (ISO, 2022).

2. Scope

This Charter applies to all information, systems, people, processes and third parties that support GHC's business operations. The programme covers information in electronic, documented, logged, backed-up and third-party-shared forms.

- Patient management systems and related healthcare data.
- Cloud platforms, infrastructure, applications, databases and networks.
- Corporate systems used by Finance, HR, Legal, Compliance, IT Operations and Development.
- Sensitive financial, employee, customer, operational and regulatory information.
- Employees, contractors, temporary workers and service providers who access GHC systems or data.
- Third-party vendors, cloud service providers, integration partners and acquired entities.
- Security policies, standards, procedures, monitoring, incident response, risk management and awareness activities.

3. Authority

The Information Security Programme is authorised by Executive Management and overseen by the Board / Risk and Audit Committee. The Director of Information Security Governance is authorised to coordinate the programme, develop policies, request risk information, escalate material risks, coordinate governance reporting and recommend risk-based security priorities.

The programme has authority to establish and maintain security policies, require business units to identify and manage security risks, request remediation plans, coordinate incident-response governance reviews and report material security matters to the CEO, Security Steering Committee and Board.

The programme does not remove operational responsibility from business units. Each department remains responsible for protecting the systems, data and processes under its control.

4. Roles and Responsibilities

Role	High-level responsibility
Board / Risk and Audit Committee	Provides oversight, approves the security governance mandate, reviews major risks and challenges management where security risk is not adequately addressed.
Chief Executive Officer	Owns executive accountability and ensures the programme has authority, visibility and support.
Director of Information Security Governance	Coordinates policy governance, risk reporting, incident readiness oversight, awareness oversight, maturity planning and Board-level reporting.
Security Steering Committee	Supports cross-functional decisions on risks, exceptions, remediation priorities, incidents, compliance concerns and security initiatives.
Chief Technology Officer	Ensures technology strategy, product development and infrastructure decisions consider security risk.

Role	High-level responsibility
IT Manager	Executes operational security activities relating to infrastructure, networks, technical controls, monitoring support and incidents.
Senior Developer	Supports secure development practices in design, development, testing, release and remediation.
Compliance Officer	Advises on healthcare regulatory obligations, data-protection requirements and compliance risk.
HR Manager	Owns onboarding, offboarding, role changes, awareness coordination and disciplinary processes linked to policy breaches.
Finance / Data Representative	Helps identify and manage risks relating to sensitive financial information and business data.

5. Key Principles

Principle	Meaning for GHC
Risk-based security	Controls and priorities must be based on risk to patients, customers, systems, operations and reputation.
Business alignment	Security must support growth, market expansion, uptime, innovation and customer trust rather than block the business unnecessarily.
Accountability	Information security is a shared responsibility; the governance function coordinates, while departments manage risks in their own areas.
Proportionate controls	Controls should be practical, affordable and strong enough for GHC's size, obligations and risk exposure.
Protection of patient trust	Patient information is central to GHC's reputation and must be protected as a priority asset.
Secure innovation	AI-driven diagnostics and secure data-exchange capabilities must include security and risk review from the start.
Continuous improvement	Security maturity, incidents, audit findings and changes in business risk must be used to improve the programme over time.

6. Reporting Structure

The Director of Information Security Governance reports to the CEO for executive direction and operational mandate and has direct reporting access to the Board / Risk and Audit Committee for material risks, significant incidents, major policy exceptions and maturity updates.

The Security Steering Committee should meet at least quarterly, or more frequently where significant risks, incidents or major projects require attention. Board reporting should use clear business language covering risk impact, likelihood, affected systems, remediation status, accountable owners and investment decisions required.

7. Review and Approval

This Charter must be approved by Executive Management and the Board / Risk and Audit Committee. It should be reviewed at least annually, or sooner following a major incident, significant regulatory change, acquisition, market expansion, major technology change or material change to GHC's risk profile.

Evidence Item 2.2 - CFO Justification Memo

To: Marcus Thorne, Chief Financial Officer

Subject: Justification for Approval of the GHC Information Security Charter

The proposed Information Security Charter gives GHC a clear mandate for managing security risk in a structured and accountable way. This matters because the organisation is growing, entering new regional markets, investing in AI-driven diagnostics and handling sensitive patient and business information. Without a formal Charter, security decisions may remain reactive, inconsistent and difficult to justify from a business or investment point of view.

From a finance and governance perspective, the Charter supports better investment decisions because it links security activity to business risk, regulatory exposure, operational resilience and customer trust. It also helps avoid uncontrolled spending by requiring security priorities to be risk-based and proportionate. Funding can therefore be directed to the areas that matter most, including patient information protection, uptime, compliance readiness and reducing the likelihood of costly incidents.

Task 3: Board Reporting and Security Metrics

Board request: Provide a concise, business-focused security update using the six-month dataset.

Evidence Item 3.1 - One-Page Board Executive Summary

Overall Security Posture

RED - immediate Board attention required

GHC's current security posture should be reported as Red because the six-month trend shows increasing security pressure while key controls are not keeping up. The organisation is experiencing more phishing attempts, more malware activity, more critical vulnerabilities, lower patching performance and a steady increase in high-risk incidents.

The most concerning point is that high-risk incidents increased every month, from 2 in September to 7 in February. At the same time, the critical patching rate fell from 67% to 55%, meaning GHC is carrying more unresolved exposure while the threat environment is becoming more active.

Although staff training improved from 45% to 70%, this has not yet reduced incidents or phishing activity. This suggests that training is moving in the right direction, but it is not enough on its own. For a health-technology organisation handling patient information, this trend creates business risk in customer trust, regulatory exposure and operational resilience.

Board Security Metrics Dashboard - September to February

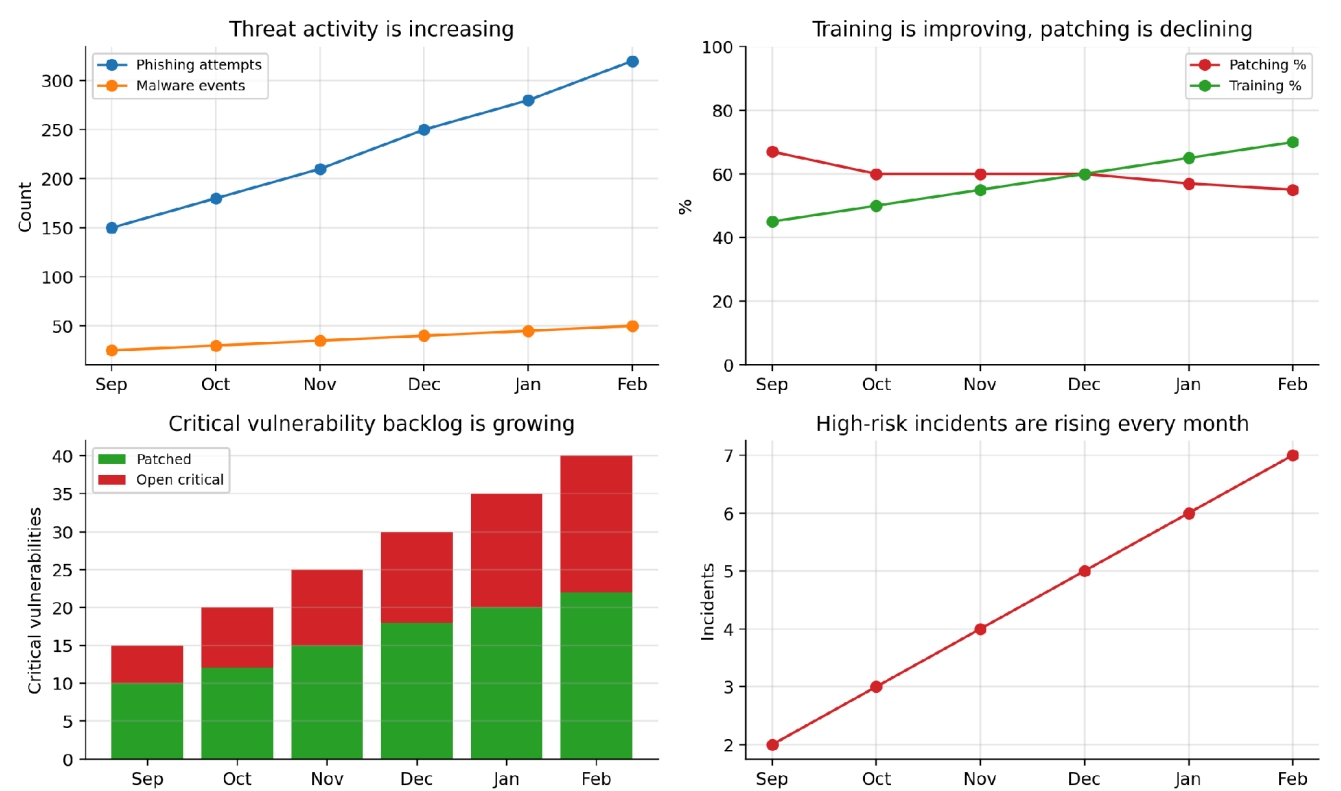


Figure 2. Board security metrics dashboard using the six-month dataset.

Evidence Item 3.2 - Five Selected Security Metrics

Metric	February status	Six-month trend	Board-level commentary
High-risk incidents	7 incidents	Increased from 2 to 7	Most serious indicator. Incidents increased every month, showing that current controls are not reducing business-impacting events.
Critical vulnerability patching rate	55%	Declined from 67% to 55%	Patching performance is getting worse while the number of critical weaknesses is increasing.
Critical vulnerabilities patched	22 of 40 patched	Open critical items increased from 5 to 18	GHC is patching more items in number, but the backlog is growing faster than remediation.
Phishing attempts	320 attempts	Increased from 150 to 320	Phishing attempts more than doubled, increasing the likelihood of account compromise, data

Metric	February status	Six-month trend	Board-level commentary
			leakage and operational disruption.
Security training completion	70%	Improved from 45% to 70%	Training is improving, but 30% of staff remain untrained and incidents are still rising.

Evidence Item 3.3 - Priority Risks and Actionable Recommendations

Priority risk / issue	Evidence from dataset	Business impact
Rising high-risk incidents	Incidents increased from 2 to 7 over six months.	Possible major breach, customer trust damage, regulatory scrutiny or service disruption.
Critical patching is falling behind	Critical vulnerabilities increased from 15 to 40 while patching fell from 67% to 55%.	Unpatched critical vulnerabilities increased from 5 to 18, creating a growing attack surface.
Training is improving but not changing outcomes yet	Training rose from 45% to 70%, but phishing and incidents still increased.	Training must be supported by practical simulations, controls and reporting processes.
Recommendation	Action required	
1. Launch a 60-day critical remediation sprint	Reduce the critical vulnerability backlog and restore critical patching performance to at least 80%. Use weekly tracking, named owners and executive escalation for delays.	
2. Strengthen phishing and malware defence	Improve phishing awareness, email filtering, suspicious email reporting, endpoint protection and incident readiness. Measure behaviour, not only training completion.	
3. Introduce monthly Board-level security reporting	Provide a simple dashboard showing incident trends, critical patching, open critical risks, phishing activity and training completion.	

Evidence Item 3.4 - Metric-Selection Rationale

The five metrics are suitable for Board oversight because they show both risk exposure and management effectiveness. High-risk incidents show actual business-impacting events. Critical vulnerability patching shows whether GHC is reducing known weaknesses. Open critical vulnerabilities show whether risk is accumulating. Phishing attempts show external pressure on the organisation. Security training completion shows whether staff are being prepared to recognise and respond to common threats.

Together, these metrics give the Board a balanced and decision-useful view. They are simple enough for executive reporting but still connected to patient trust, regulatory risk, operational resilience and investment priorities.

Task 4: Security Steering Committee

CEO request: Establish a formal Security Steering Committee to resolve cross-functional issues, including the dispute between the CTO and IT Manager regarding password policy.

Evidence Item 4.1 - Security Steering Committee Terms of Reference

TOR section	Terms of Reference
Purpose	The purpose of the Security Steering Committee is to provide a formal cross-functional forum for making information security decisions that affect more than one department. The committee will ensure that decisions are risk-based, practical, business-aligned and properly escalated where required.
Scope	The committee will consider organisation-wide or cross-functional matters including policies, major control changes, risks affecting patient information, security requirements for product development, incident readiness, policy exceptions, remediation priorities, compliance issues, acquisition risks, vendor risks and awareness concerns.
Membership	Membership should include the Director of Information Security Governance as Chair, the CTO, CFO or delegate, IT Manager, Senior Developer, Compliance Officer, HR Manager and Finance / Data Representative. Subject-matter experts may be invited where required.
Responsibilities	Responsibilities include reviewing major security risks, recommending policies and standards, resolving cross-functional disputes, reviewing incidents and near-misses, monitoring remediation progress, reviewing exceptions and escalating material risks.
Meeting cadence	The committee should meet monthly for the first six months while the governance programme is being formalised. It may then move to quarterly if risk levels are stable. Extraordinary meetings may be called for major incidents, urgent conflicts or regulatory concerns.
Decision authority	The committee may recommend and endorse governance decisions, agree remediation priorities and resolve management-level disagreements. It should not approve material risk acceptance alone. Material risks must be escalated to the CEO and, where appropriate, the Board.
Reporting	The Director of Information Security Governance will prepare a short report after each meeting covering decisions, risks reviewed, actions, owners, due dates, escalations and matters for CEO or Board attention.

Evidence Item 4.2 - Sample Agenda for First SSC Meeting

Agenda item	Lead	Time
1. Welcome and purpose of the committee	Director of Information Security Governance	5 min
2. Review of GHC context: growth, acquisitions and near-miss	Director of Information Security Governance	10 min
3. Confirm membership, responsibilities and decision process	Director of Information Security Governance	10 min
4. Password-policy conflict: IT Manager's concern	IT Manager	10 min
5. Password-policy conflict: CTO's concern	CTO	10 min
6. Governance assessment of the proposed policy	Director of Information Security Governance	15 min
7. Agree balanced password and authentication approach	Committee discussion	20 min
8. Confirm implementation actions, owners and timelines	Chair and members	5 min
9. Confirm reporting and escalation requirements	Chair	5 min

Proposed Password-Policy Decision

The committee should not approve the proposed policy in its current form. A 16-character complex password changed every 30 days without password-manager integration is likely to create unnecessary friction and may encourage users to choose predictable patterns, write passwords down or reuse passwords.

The recommended position is to approve a more balanced authentication standard based on strong password length, screening against commonly used or compromised passwords, multi-factor authentication for sensitive and privileged access, password-manager support, password changes when there is evidence of compromise, stronger monitoring and higher controls for privileged accounts. This is aligned with NIST SP 800-63B-4 guidance that verifiers should not impose composition rules or require periodic password changes, and should allow password managers and autofill (Temoshok et al., 2025).

Evidence Item 4.3 - Briefing Note to the CEO

To: Sarah Chen, Chief Executive Officer

Subject: Establishing the Security Steering Committee

The Security Steering Committee will give GHC a structured way to resolve security decisions that affect multiple departments. The current password-policy dispute shows why this is needed. The IT Manager is correctly focused on stronger controls, while the CTO is correctly concerned about user friction, developer productivity and practical implementation. Without a formal committee, these issues can become personal or departmental disagreements instead of risk-based business decisions.

The committee will help maintain strategic alignment by bringing the right people into the same decision forum. Security decisions will be reviewed against business impact, patient trust, regulatory obligations, cost, uptime and innovation. Where a decision creates material risk or requires executive approval, the committee will escalate it clearly to the CEO and Board rather than allowing unresolved conflicts to delay action.

Task 5: Governance Maturity Assessment

Board request: Assess current maturity and propose a realistic roadmap to reach at least Level 3 - Defined across all governance domains within 12-18 months.

Evidence Item 5.1 - Completed Maturity Assessment

GHC's current maturity is low to moderate. Some security-related activities exist, but they are not consistent, fully documented or clearly owned. This is expected in an organisation that has grown quickly and absorbed acquisitions, but it creates a risk that security is being managed reactively rather than through a formal governance programme.

Governance domain	Current-state evidence	Current level	Assessment summary
Policy & Documentation	Policies are inconsistent, inherited from acquisitions and not always current.	Level 2 - Initial	Some policies exist, but they are not standardised or reliably maintained. GHC needs a single approved policy framework.
Roles & Responsibilities	Security ownership and accountability are unclear.	Level 1 - Ad Hoc	This is one of the weakest areas. Without clear ownership, decisions and escalation may be inconsistent.
Risk Management	The organisation is reactive and lacks a formal proactive assessment process.	Level 1 - Ad Hoc	Risk management is not embedded into normal business planning. Risks are likely being addressed after problems arise.
Metrics & Reporting	Technical metrics exist, but they are not consistently translated into business impact.	Level 2 - Initial	Security data exists, but it is not yet mature enough for consistent Board-level decision-making.
Training & Awareness	Annual training is mandatory, but engagement is low.	Level 2 - Initial	Training exists, but it is treated mainly as a compliance activity rather than a behaviour-change programme.
Compliance	Preparation is audit-driven rather than continuously monitored.	Level 2 - Initial	Compliance activity happens, but it is too reactive and focused on audit preparation instead of ongoing readiness.

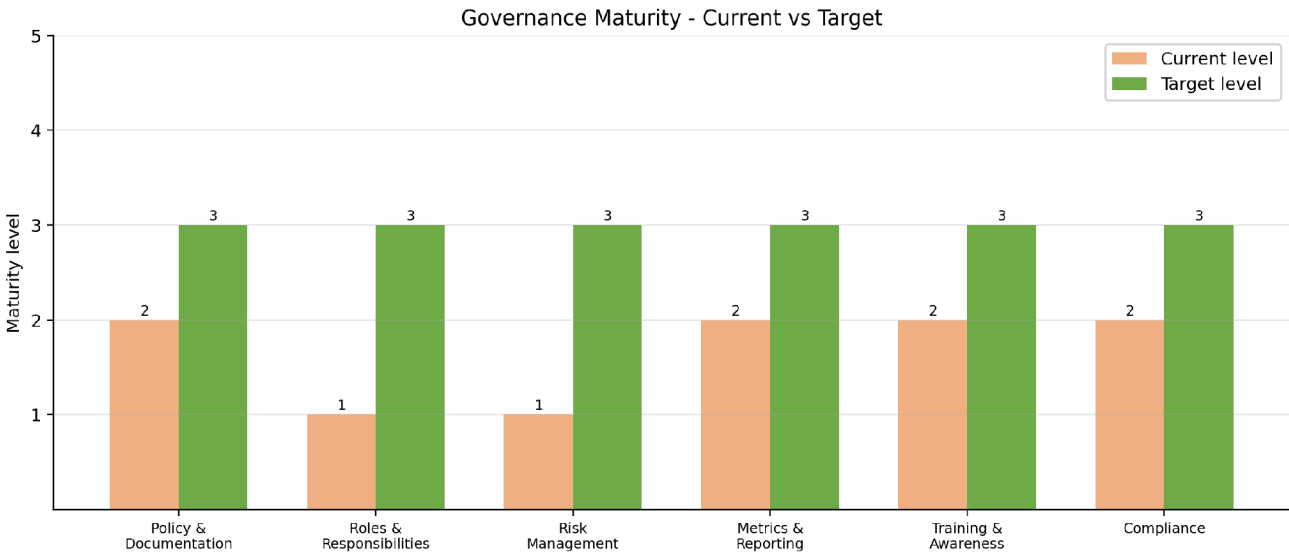


Figure 3. Current maturity against the Level 3 target.

Evidence Item 5.2 - 12-18 Month Maturity Roadmap

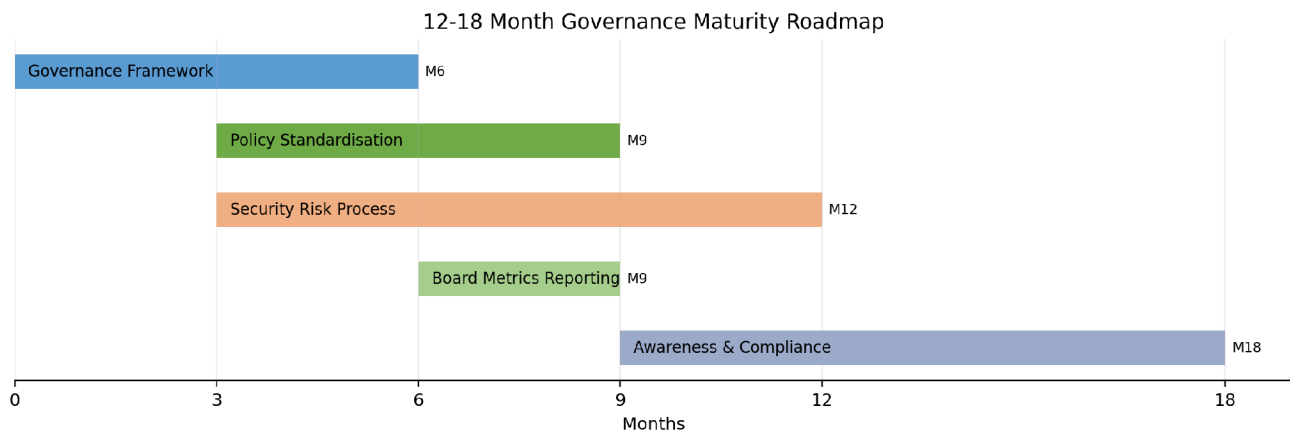


Figure 4. 12-18 month maturity roadmap timeline.

Initiative 1: Establish the Security Governance Framework

Area	Detail
Objective	Create a formal governance structure with clear authority, accountability and decision-making.
Key activities	Approve the Information Security Charter; establish the Security Steering Committee; define reporting lines to the CEO and Board; confirm governance roles and responsibilities; implement the RACI matrix.
Measurable expected outcome	By month 6, the Charter is approved, the SSC is operating, governance roles are documented and major security decisions have an accountable owner.
Target maturity impact	Roles & Responsibilities move from Level 1 to Level 3.

Initiative 2: Standardise the Policy and Documentation Framework

Area	Detail
Objective	Replace inconsistent and inherited policies with a single GHC-approved security policy framework.
Key activities	Review policies from GHC and acquired entities; identify gaps and duplicates; approve a core policy set; assign policy owners; introduce annual review cycles and version control.
Measurable expected outcome	By month 9, all core policies are approved, current, centrally stored, assigned to owners and communicated to relevant staff.
Target maturity impact	Policy & Documentation move from Level 2 to Level 3.

Initiative 3: Implement a Proactive Security Risk Management Process

Area	Detail
Objective	Move GHC from reactive issue handling to proactive identification, assessment and treatment of security risks.
Key activities	Create a security risk register; define risk rating criteria; perform quarterly risk reviews; assign risk owners; track treatment actions; escalate material risks to the CEO and Board.
Measurable expected outcome	By month 12, GHC has an active risk register, quarterly risk reviews, named owners and documented treatment plans for priority risks.
Target maturity impact	Risk Management moves from Level 1 to Level 3.

Initiative 4: Develop Board-Level Security Metrics and Reporting

Area	Detail
Objective	Convert technical security data into clear business reporting for executive and Board oversight.
Key activities	Select Board-level metrics; define reporting frequency; agree Red/Amber/Green thresholds; include business impact commentary; report trends and required decisions.
Measurable expected outcome	By month 9, the Board receives a consistent dashboard covering risk posture, incidents, vulnerability remediation, training and compliance readiness.
Target maturity impact	Metrics & Reporting move from Level 2 to Level 3.

Initiative 5: Strengthen Security Awareness and Continuous Compliance Readiness

Area	Detail
Objective	Improve staff engagement and move compliance from audit-driven preparation to continuous readiness.
Key activities	Refresh annual training; introduce role-based awareness for HR, Finance, Development and IT; run phishing awareness activities; track completion and engagement; create a compliance obligations register; perform periodic internal checks.
Measurable expected outcome	By month 18, training reaches at least 90%, role-based training is in place for high-risk teams and compliance evidence is maintained throughout the year.
Target maturity impact	Training & Awareness and Compliance move from Level 2 to Level 3.

Evidence Item 5.3 - Board Executive Summary

GHC's current information security governance maturity is between Level 1 - Ad Hoc and Level 2 - Initial. Some activities exist, such as policies, training, compliance preparation and technical metrics, but they are not yet consistent, fully documented, clearly owned or regularly translated into business risk. This position is not unusual for an organisation that has grown quickly and absorbed acquisitions, but it is not suitable for GHC's current risk profile as a health-technology provider handling sensitive patient information.

The proposed 12-18 month roadmap is strategically important because it gives GHC a realistic path to reach Level 3 - Defined across all key governance domains. The focus is not to over-engineer security, but to create clear ownership, approved policies, proactive risk management, meaningful Board reporting, better staff awareness and continuous compliance readiness. This will support GHC's growth, protect customer trust, improve accountability and reduce the likelihood that future security issues become major business incidents. A defined risk-management lifecycle also supports recognised risk-management concepts of accountability, continuous monitoring and treatment tracking (Joint Task Force, 2018).

References

- International Organization for Standardization. (2022). ISO/IEC 27001:2022 - Information security management systems. ISO. <https://www.iso.org/standard/27001.html>
- Joint Task Force. (2018). Risk Management Framework for Information Systems and Organizations: A System Life Cycle Approach for Security and Privacy (NIST Special Publication 800-37 Rev. 2). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.SP.800-37r2>
- National Institute of Standards and Technology. (2024). The NIST Cybersecurity Framework (CSF) 2.0 (NIST CSWP 29). <https://doi.org/10.6028/NIST.CSWP.29>
- Temoshok, D., Fenton, J., Choong, Y.-Y., Lefkovitz, N., Regenscheid, A., Galluzzo, R., & Richer, J. (2025). Digital Identity Guidelines: Authentication and Authenticator Management (NIST Special Publication 800-63B-4). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.SP.800-63b-4>

AI Assistance Declaration

AI tools were used to support brainstorming, structuring, and language refinement. The student reviewed and adapted the content and remains responsible for the accuracy, reasoning, and final submission.